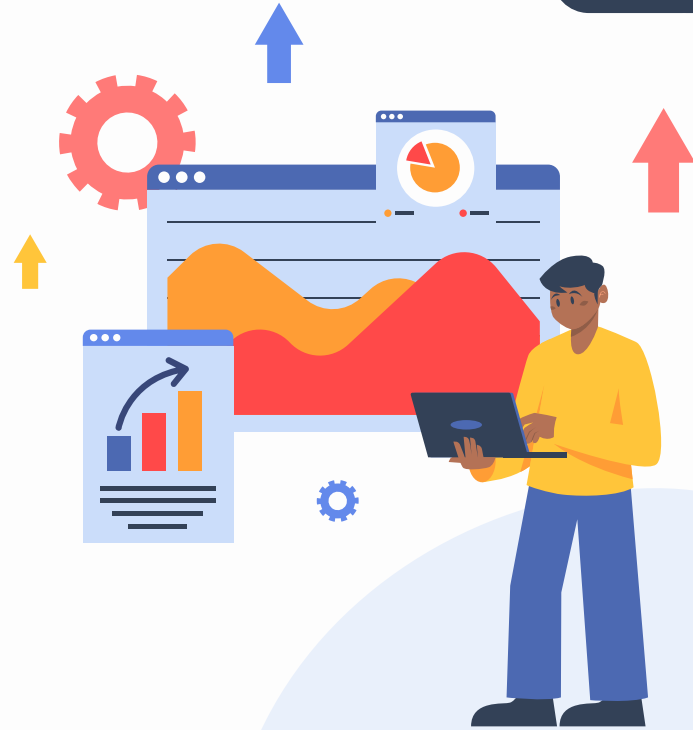


Impact of Weather and Airport Volume on Flights

Authors: Emanuel Pacheco, Ivo Simões, Filipe Obrist



Motivation & Objective

Volatile factors (weather, structural constraints), coupled with complex boarding processes, contribute to overall **systemic inefficiency** in aviation.

We investigate the influence of **weather** conditions and **passenger volume** on airline operations (**delays/cancellations**) and market behavior (ticket **prices**).





Core Research Questions

Weather Impact

Does weather significantly affect travel time, cancellations or delays in specific locations?

Volume Correlation

Are delays related to TSA passenger volumes?

Market Dynamics

Does location or number of flights influence average ticket pricing?

Seasonality

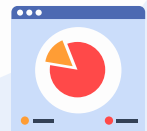
How do ticket prices vary between "stormy" and "clear" quarters? Do high prices influence cancellations?

Predictive

Forecasting future pricing based on historical patterns

Prescriptive

Identifying optimal travel windows to minimize risk





Sources & Datasets



BTS On-Time Performance

Individual flight records
(25 millions records)



BTS Air Carrier Statistics

Market data and
passenger volumes
(15 thousands records)

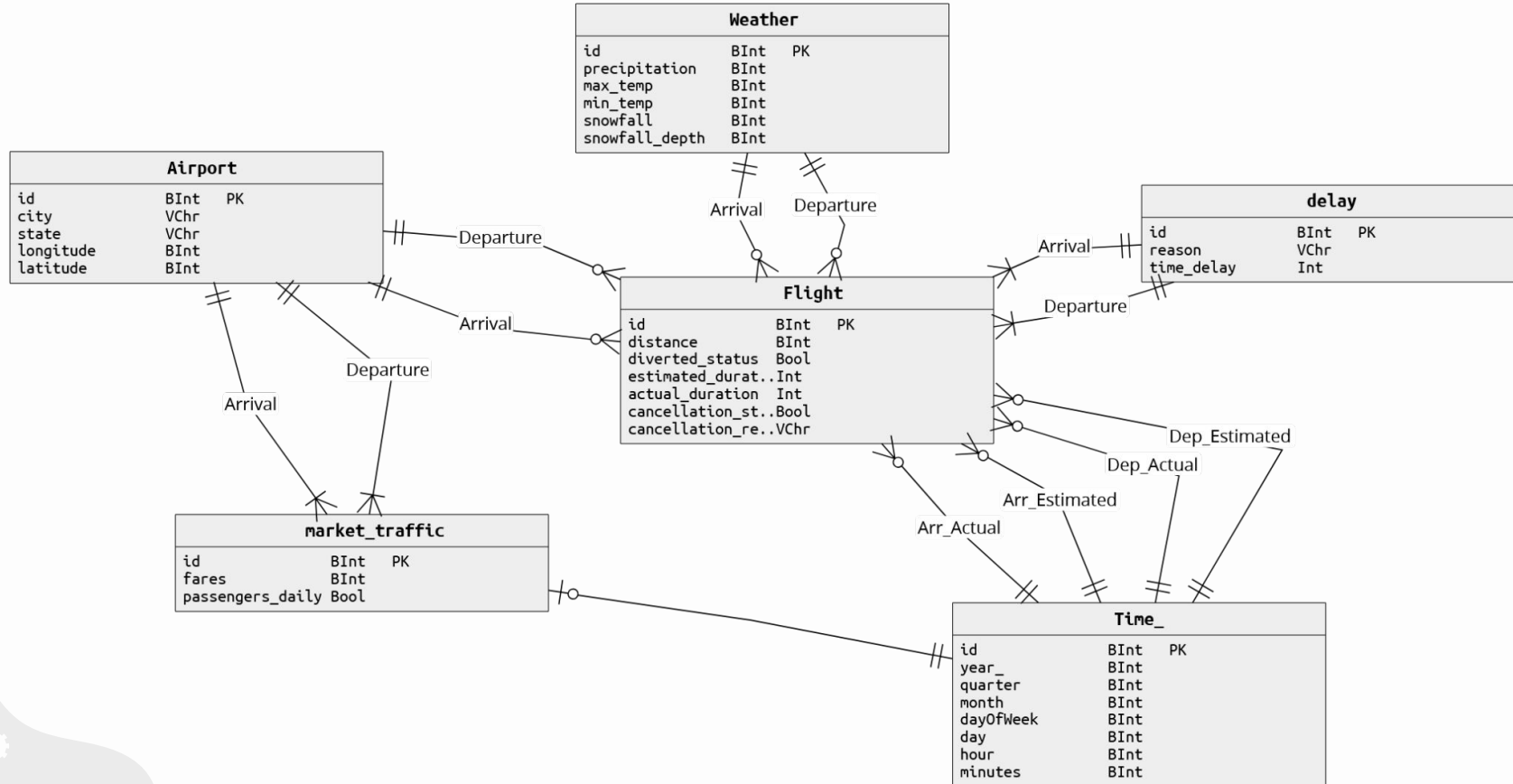


NOAA Integrated Surface Dataset

Global weather
observations
(28 million records)



Star Schema Design



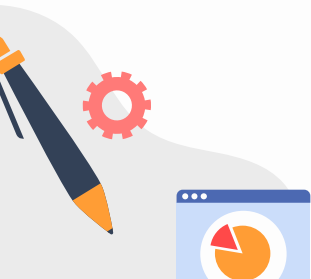
ETL: Initial vs. Improved

Initial Approach:

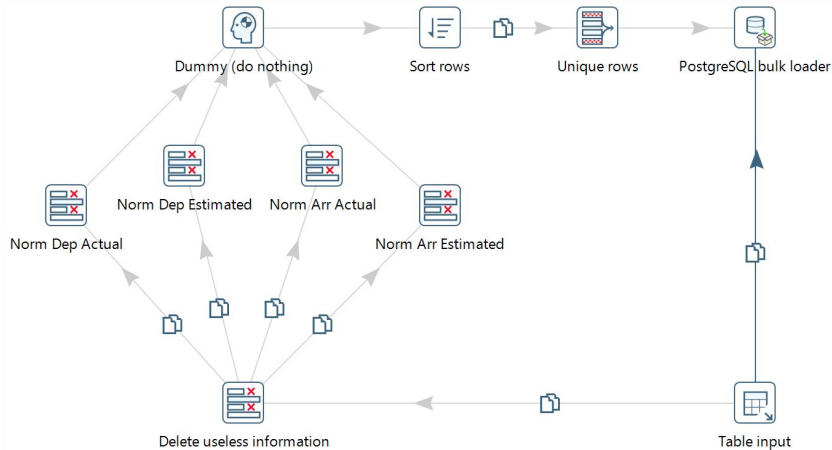
- We executed everything inside Pentaho (e.g., row filtering, lookups, data merging);
- Data was constantly passed from transformation to transformation via files, causing massive I/O delays;
- Indexing transformations became a critical bottleneck within the pipeline;
- Uploading just one year of data took over 8 hours.

Improved Approach:

- We created three temporary tables in our PostgreSQL database to store the raw, clean data;
- We shifted the heavy merging and indexing logic out of Pentaho into SQL queries;
- Leveraged PostgreSQL, which is optimally perfect for relational joins and index solving;
- We processed the entire 4 years of data in just 45 minutes (98% Improvement).



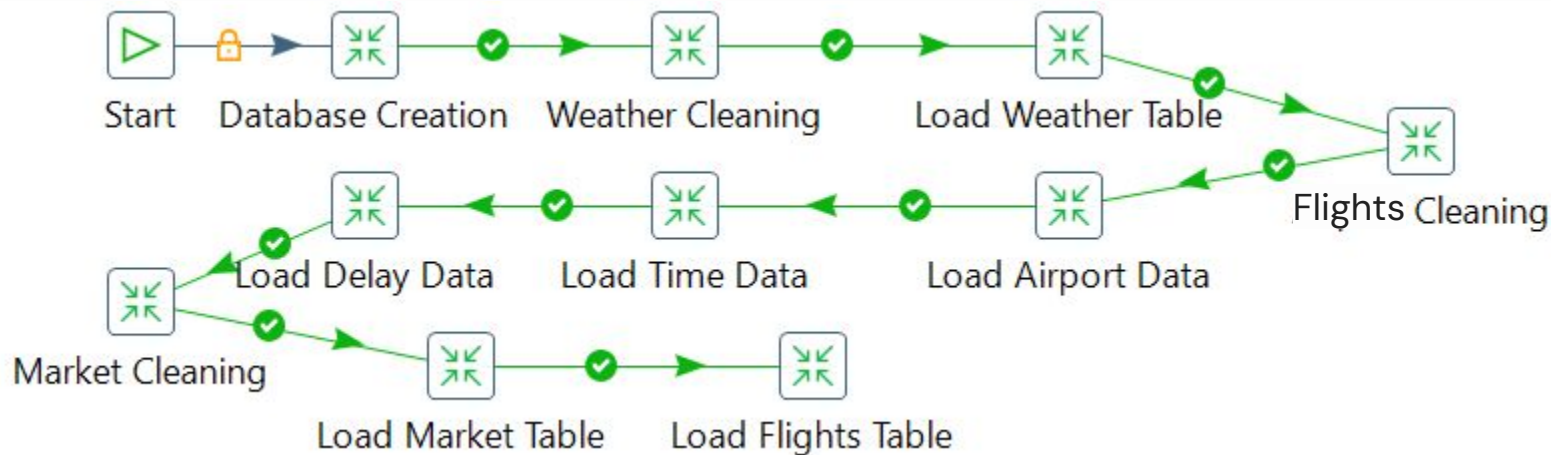
ETL: Workflow (Example)



```
SELECT DISTINCT
    year_, quarter, "month", "day", dayofweek,
    t.hour, t.minute
FROM "public".flights_temporary,
LATERAL (
    VALUES
        (dep_actual_hour, dep_actual_minute),
        (dep_estimated_hour, dep_estimated_minute),
        (arr_actual_hour, arr_actual_minute),
        (arr_estimated_hour, arr_estimated_minute)
) AS t(hour, minute)
ORDER BY year_, "month", "day", hour, minute;
```



ETL: Final Workflow



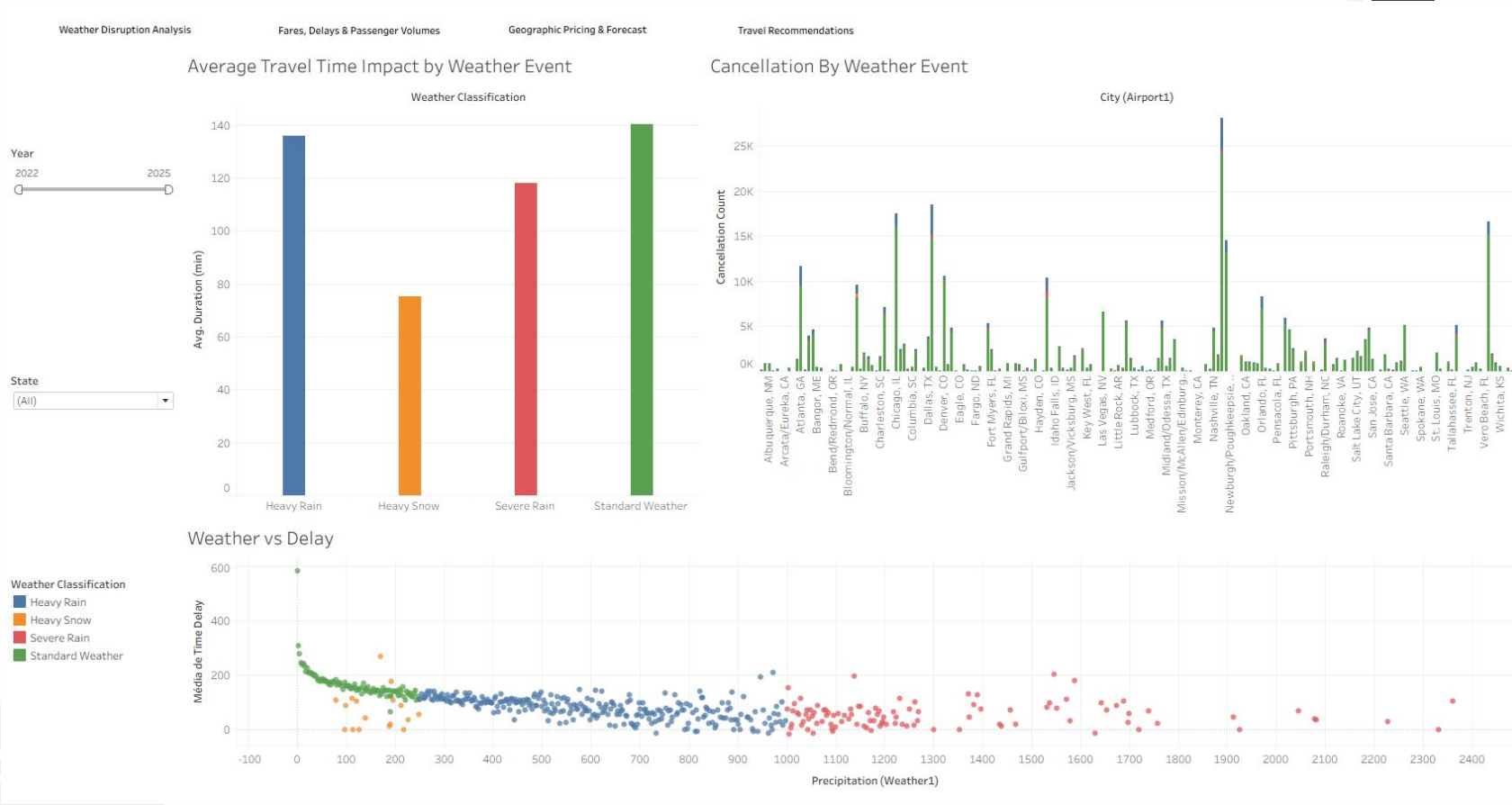


ETL: Final Table Statistics

Table	Estimated Rows	Data Size	Index Size	Total Size
Flight	20,360,852	2744 MB	436 MB	3180 MB
Time	1,929,157	172 MB	41 MB	213 MB
Weather	1,718,053	126 MB	37 MB	163 MB
Market Traffic	52,420	3952 kB	1280 kB	5232 kB
Delay	22,225	1528 kB	504 kB	2032 kB
Airport	196	48 kB	16 kB	64 kB



OLAP Analysis (Tableau): Weather Analysis



OLAP Analysis (Tableau): Market & Volume



Weather Disruption Analysis

Fares, Delays & Passenger Volumes

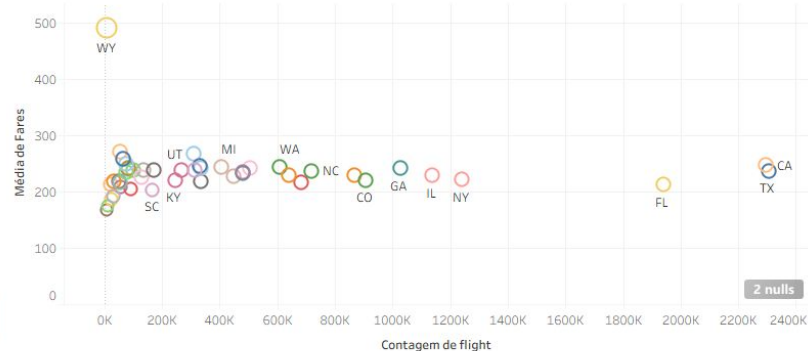
Geographic Pricing & Forecast

Travel Recommendations

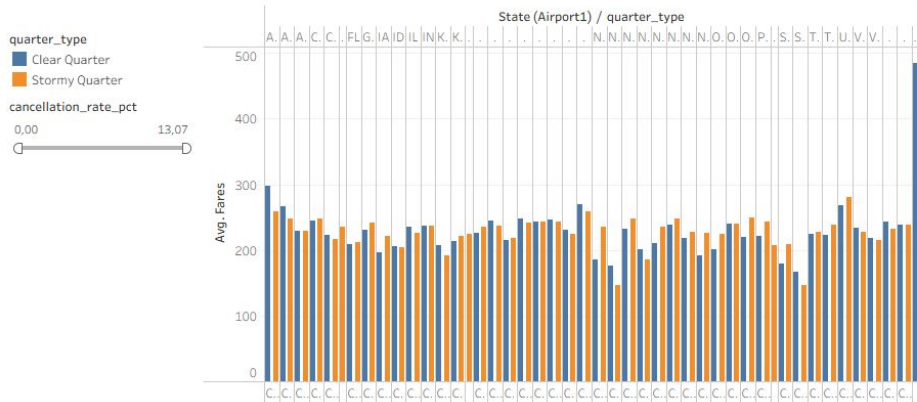
Flights vs Passengers



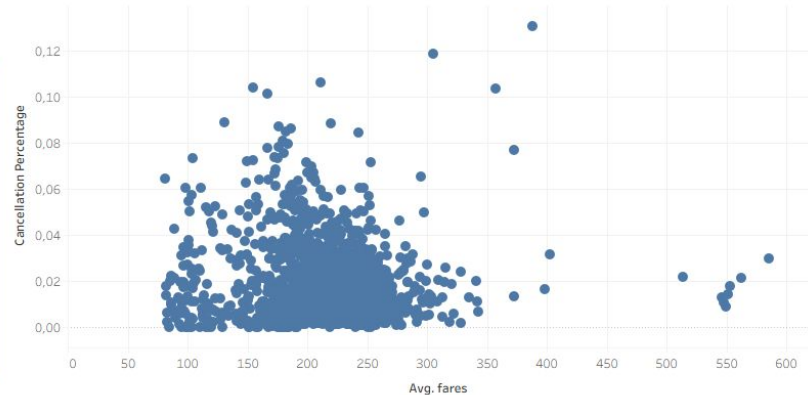
Flights vs Fare By State



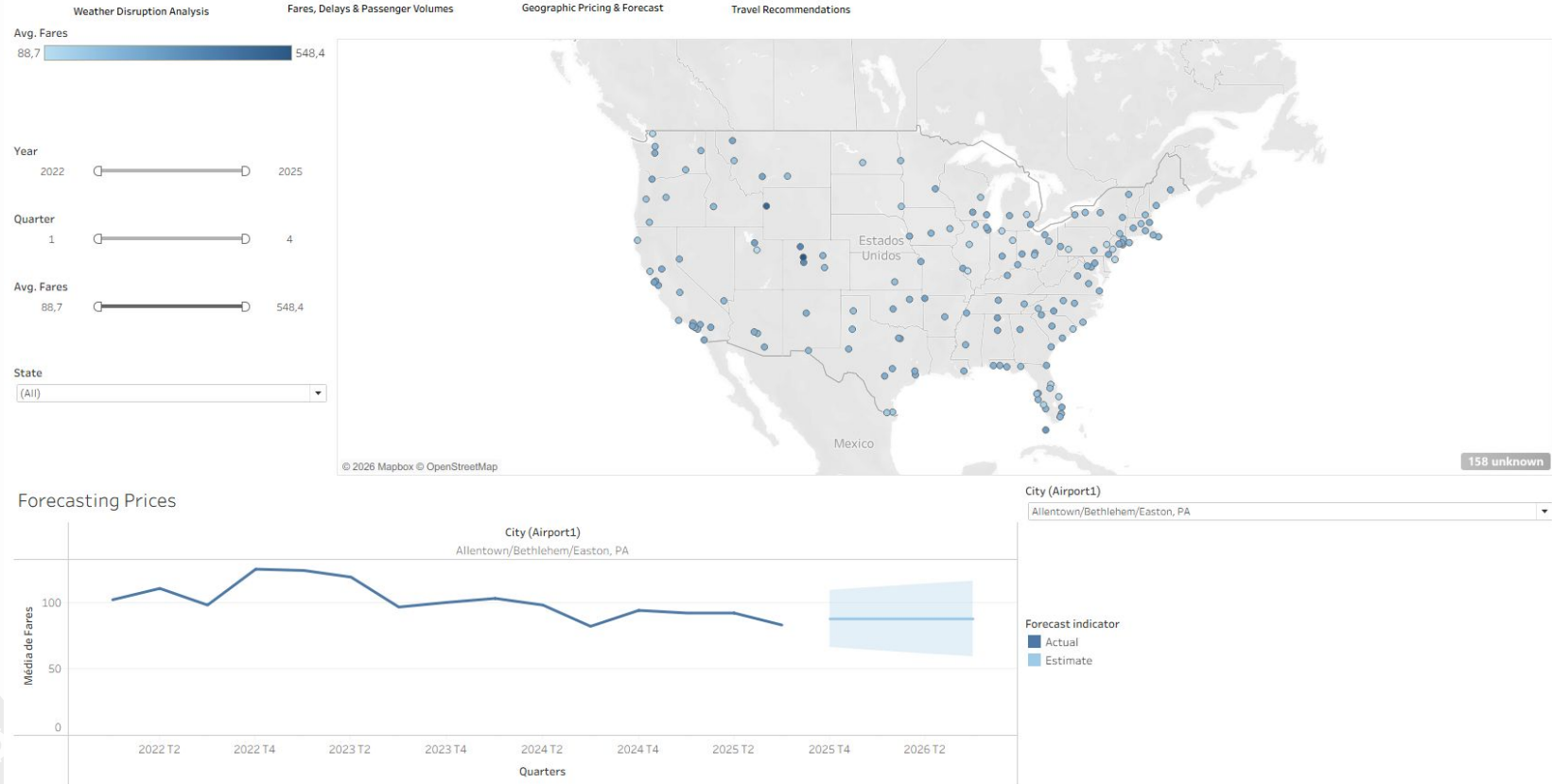
Fares vs. Stormy/Clear Quarters



Fares vs. Cancellation



OLAP Analysis (Tableau): Geo-Pricing & Forecast



OLAP Analysis (Tableau): Travel Recommendations



Weather Disruption Analysis

Fares, Delays & Passenger Volumes

Geographic Pricing & Forecast

Travel Recommendations

Quarters to Travel

State (Airport1)

(Multiple values)

City (Airport1)

(Multiple values)

Avg. Fares

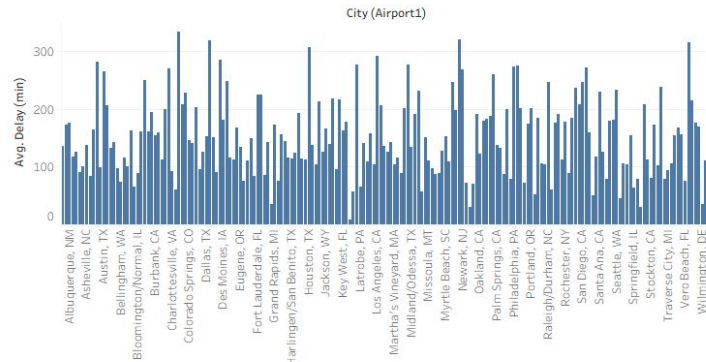


City to Travel (Delay)

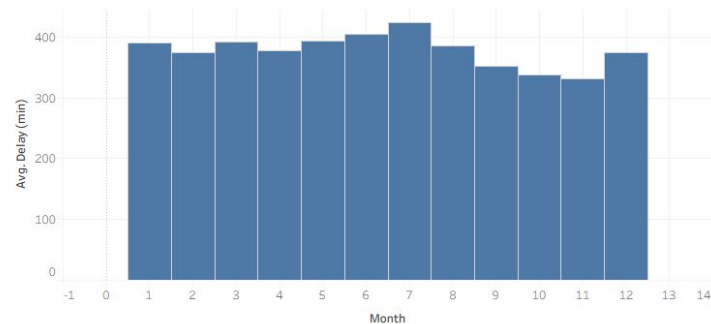
Year (Time 1)



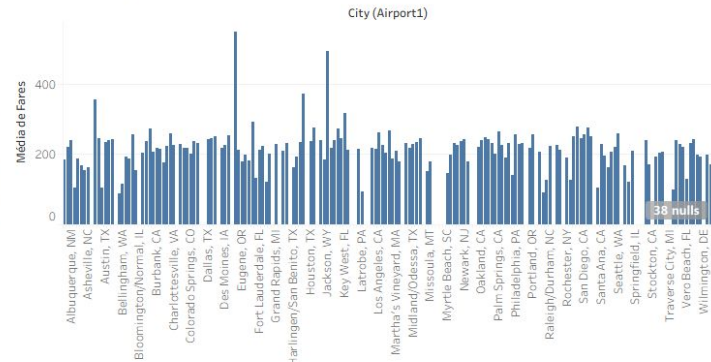
Avg. Delay (min)



Month to Travel



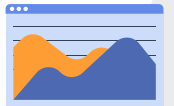
City to Travel (Costs)





Results & Comments

- **Weather Findings:** Weather adversities can cause localized "spikes" in travel time and drive **cancellations**, rather than just consistent delays, in certain places.
- **Volume Findings:** There is a moderate negative correlation between **TSA passenger counts** and **delays**, indicating that airport congestion doesn't decrease throughput.
- **Market Resilience:** Despite the volatile factors mentioned above, the market remained **resilient** across the studied period, even during quarters with higher average fares.





Conclusions

Technical Success

Successfully architected a pipeline capable of processing 20M+ rows with good performance.

Analytical Takeaway

The system shows significant resilience to passenger volume and weather, regardless of price volatility.

Future Work

Increase data granularity for more precise analysis and forecasting. Expand to other territories.



Thanks!

Do you have any questions?

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